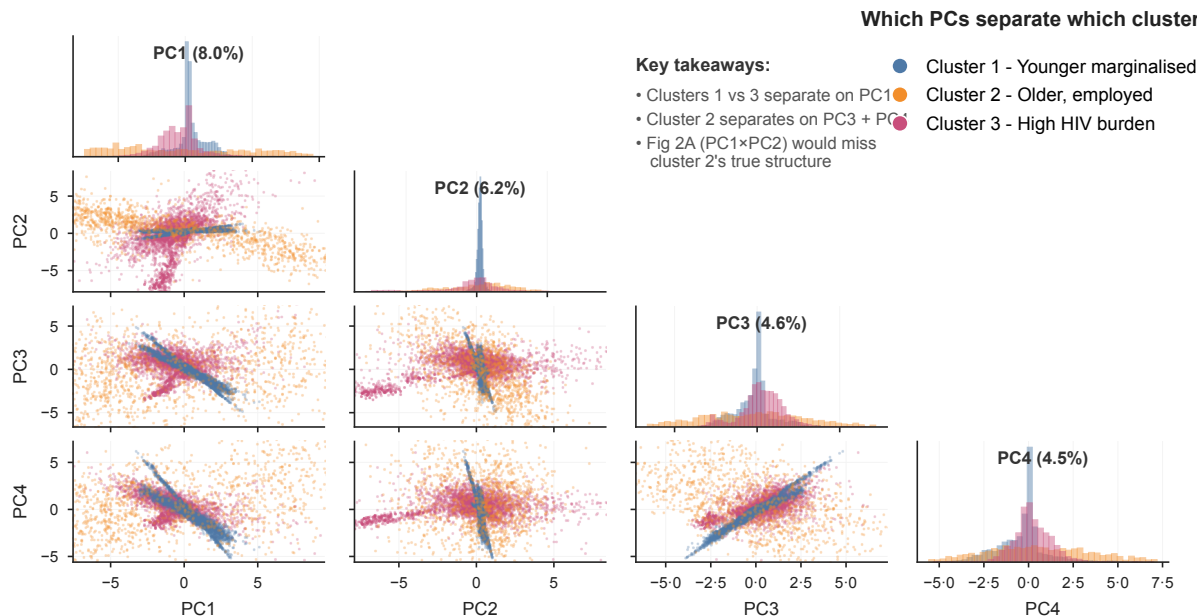


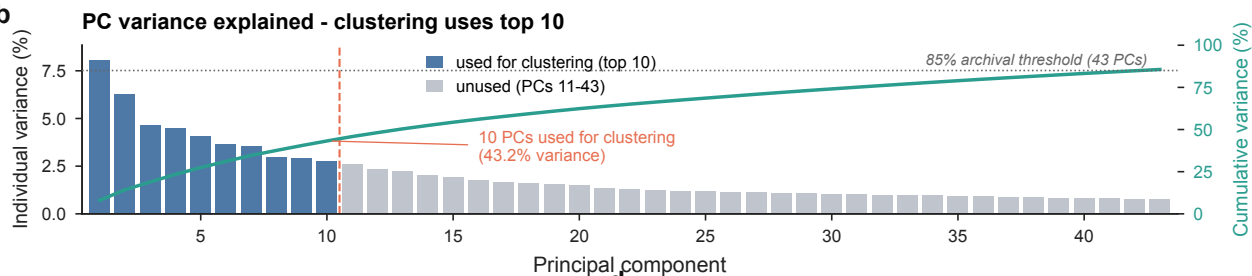
The PC1×PC2 projection in Figure 2A captures 14.3% of the variance on which clustering is performed. Panel a shows where clusters actually separate across PC1-4; b-d confirm dimensionality, separation quality, and cluster tightness in the 10D space GMM uses.

a



Overall silhouette: GMM-k3 = 0.09 • k-means-k3 = 0.27 • GMM ARI stability (10D) = 0.985

b



c

